

Unit 3 - Classical Encryption Techniques

Cryptography and Web Security (DI03016021) - GTU Diploma IT Sem 3

26% weightage | 12 hours (BIGGEST unit - give it the most time)

How to use these notes: each topic is written the way you can copy it in the exam. For a **3-mark** question write only the definition + 2-3 points. For **4 marks** add an example or table. For **7 marks** write everything, with diagram and table. Lines marked ★ show where the topic was already asked in a GTU paper (only 2 papers exist so far: Winter 2025 and Summer 2026).

Questions already asked from this unit (do these first)

- Asked in S2026 Q1(b) [4M: differentiate substitution and transposition with example]
- Additive cipher asked in BOTH papers: W2025 Q1(b) [4M] + S2026 Q3(a)OR [3M]
- Multiplicative cipher: W2025 Q5(a) [3M numerical]
- Asked in S2026 Q3(b) [4M: affine HELLO $a=3$, $b=7$, encrypt and decrypt]
- Asked in W2025 Q3(b) [4M: Autokey with example]
- Asked in S2026 Q3(a) [3M: Vigenère KEEPMESSAGESECRET key CIPHER]
- Asked in W2025 Q3(b)OR [4M: Playfair, plaintext battery, key FORUM]
- Hill cipher asked in BOTH papers: W2025 Q3(c) [7M: decisions, 3×3] + S2026 Q3(b)OR [4M: DATA, 2×2]
- Asked in W2025 Q5(b)OR [4M: one-time pad with example]
- Asked in W2025 Q3(a)OR [3M: keyed transposition with example]
- Asked in S2026 Q3(c) [7M: keyed columnar, MEETMEAFETERTHEEXAM, key 4312]
- Asked in W2025 Q3(a) [3M: stream cipher and block cipher with example]
- RSA asked in BOTH papers (7M each): W2025 Q3(c)OR [key generation + enc/dec with example] + S2026 Q3(c)OR [$p=5$, $q=11$, $e=7$, $M=4$]

3.1 Symmetric Cipher Model



The **same secret key** is used in both the encryption and decryption boxes (symmetric = single key).

A symmetric cipher has **five ingredients**:

1. **Plaintext** - original readable message.
2. **Encryption algorithm** - performs substitutions/transpositions on the plaintext.
3. **Secret key** - input to the algorithm; exact result depends on it.
4. **Ciphertext** - scrambled message produced.
5. **Decryption algorithm** - encryption run in reverse, using the same key, to recover plaintext.

Two requirements: a **strong algorithm** and a **secret key shared safely** between sender and receiver. Mathematically: $C = E(K, P)$ and $P = D(K, C)$.

3.1.2-3.1.3 Cryptanalysis, types of attacks, brute force

Cryptanalysis = trying to break a cipher. Types of attacks depend on what the attacker knows:

Attack type	What attacker knows / can do
Ciphertext-only	Only the ciphertext and the algorithm. Hardest for attacker.
Known-plaintext	Some plaintext-ciphertext pairs as well.
Chosen-plaintext	Attacker can choose plaintext and get its ciphertext.
Chosen-ciphertext	Attacker can choose ciphertext and get its decryption.

Brute-force attack: try **every possible key** one by one until a readable plaintext appears.

Effort grows with key size: an additive cipher has only 25 useful keys (broken in seconds), while a 128-bit key has 2^{128} keys (practically impossible). A cipher is **computationally secure** if the cost of breaking it is more than the value of the data or takes longer than the useful life of the data.

Substitution vs Transposition

★ Asked in S2026 Q1(b) [4M: differentiate substitution and transposition with example]

Point	Substitution technique	Transposition technique
Meaning	Each letter is replaced by another letter / number / symbol	Positions of letters are rearranged ; letters stay the same
Identity of letters	Changed	Not changed (only order changes)
Types	Additive, multiplicative, affine, monoalphabetic, Vigenère, Playfair, Hill, one-time pad	Rail fence, rectangular (columnar), keyed transposition, keyed columnar
Example	HELLO → KHOOR (Caesar, shift 3)	HELLO → HLOEL (rail fence, 2 rails)
Weakness	Letter-frequency analysis	Anagramming; letter frequency stays same as plaintext

Modern ciphers (DES, AES) combine **both** techniques repeatedly.

3.2.1 Additive (Caesar) cipher

★ Additive cipher asked in BOTH papers: W2025 Q1(b) [4M] + S2026 Q3(a)OR [3M]

Additive cipher (Caesar / shift cipher): letters are numbered **a=0, b=1 ... z=25**. Key K is added to each letter, everything mod 26.

Encryption: $C = (P + K) \bmod 26$ Decryption: $P = (C - K) \bmod 26$

Key space is only 26 (25 useful keys), so it is easily broken by **brute force** - try all 25 keys. Caesar cipher is the case $K = 3$.

Solved (W2025): plaintext "transform", key = 9

Plain letter	T	R	A	N	S	F	O	R	M
P (number)	19	17	0	13	18	5	14	17	12
$(P + 9) \bmod 26$	2	0	9	22	1	14	23	0	21
Cipher letter	C	A	J	W	B	O	X	A	V

Ciphertext = CAJWBOXAV

Example of one step: $t = 19 \rightarrow (19 + 9) \bmod 26 = 28 \bmod 26 = 2 \rightarrow C$. For $r = 17$: $(17+9) = 26 \bmod 26 = 0 \rightarrow A$.

Solved (S2026): plaintext "SECURITY", key = 5

Plain letter	S	E	C	U	R	I	T	Y
P (number)	18	4	2	20	17	8	19	24
(P + 5) mod 26	23	9	7	25	22	13	24	3
Cipher letter	X	J	H	Z	W	N	Y	D

Ciphertext = XJHZWNYD

3.2.1 Multiplicative cipher

★ **Multiplicative cipher: W2025 Q5(a) [3M numerical]**

Multiplicative cipher: key K is multiplied with plaintext number. K must be **co-prime with 26** (so an inverse exists).

Encryption: $C = (P \times K) \bmod 26$ Decryption: $P = (C \times K^{-1}) \bmod 26$

Valid keys: **1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25** (12 keys). Inverse pairs: 3↔9, 5↔21, 7↔15, 11↔19, 17↔23; 1, 25 are their own inverses.

Solved (W2025): decrypt "dagcqo", key = 11, inverse = 19

Decryption: $P = (C \times 19) \bmod 26$. (Check: $11 \times 19 = 209 = 8 \times 26 + 1$ ✓)

Cipher letter	D	A	G	C	Q	O
C (number)	3	0	6	2	16	14
$C \times 19$	57	0	114	38	304	266
$(C \times 19) \bmod 26$	5	0	10	12	18	6
Plain letter	f	a	k	m	s	g

Plaintext = fakmsg

Verification: encrypt $f = 5 \rightarrow 5 \times 11 = 55 \bmod 26 = 3 = d$ ✓.

3.2.1 Affine cipher

★ **Asked in S2026 Q3(b) [4M: affine HELLO a=3, b=7, encrypt and decrypt]**

Affine cipher = multiplicative + additive cipher together. Two keys (a, b); **a must be co-prime with 26**.

Encryption: $C = (a \times P + b) \bmod 26$ Decryption: $P = a^{-1} \times (C - b) \bmod 26$

Encryption (a = 3, b = 7)

Plain letter	H	E	L	L	O
P	7	4	11	11	14
$3P + 7$	28	19	40	40	49
$\bmod 26 = C$	2	19	14	14	23
Cipher letter	C	T	O	O	X

Ciphertext = CTOOX

Decryption

Inverse of $a = 3$ in Z26 is **9** because $3 \times 9 = 27 = 1 \times 26 + 1$.

Cipher letter	C	T	O	O	X
C	2	19	14	14	23
$9 \times (C - 7)$	-45	108	63	63	144
$\text{mod } 26 = P$	7	4	11	11	14
Plain letter	H	E	L	L	O

Plaintext = HELLO ✓

Number of keys = $12 \times 26 = 312$.

3.2.1 Monoalphabetic substitution cipher

Monoalphabetic substitution cipher: every plaintext letter is replaced by **one fixed** other letter, using a key which is a **shuffled alphabet** (permutation of 26 letters). Number of keys = $26! \approx 4 \times 10^{26}$, so brute force is impractical, but it is broken by **letter-frequency analysis** (E, T, A are most common in English).

Plain	a b c d e f g h i j k l m	n o p q r s t u v w x y z
Cipher	Q W E R T Y U I O P A S D	F G H J K L Z X C V B N M

Example: **HELLO → ITSSG**. Additive, multiplicative and affine ciphers are special (smaller-key) cases of monoalphabetic substitution.

3.2.2 Autokey cipher

★ Asked in W2025 Q3(b) [4M: Autokey with example]

Autokey cipher is a polyalphabetic cipher. The key is **one starting key + the plaintext itself**. So the key stream never repeats, which makes it stronger than Vigenère.

Encryption: $C_i = (P_i + k_i) \text{ mod } 26$, where $k_1 = \text{given key}$ and $k_i = P_{i-1}$ for $i > 1$.

Example: plaintext HELLO, key = 7 (letter H)

Plain letter	H	E	L	L	O
P	7	4	11	11	14
Key stream	7	7	4	11	11
$(P + \text{key}) \text{ mod } 26$	14	11	15	22	25
Cipher letter	O	L	P	W	Z

Key stream = 7, then H(7), E(4), L(11), L(11). **Ciphertext = OLPWZ**.

Decryption: $P = (C - k) \text{ mod } 26$. First letter uses the given key; each decrypted plaintext letter becomes the next key. $O - 7 = 7 = H$; then $L - H(7) = 11 - 7 = 4 = E$; and so on.

Weakness: the key stream contains the plaintext, so attacks guessing words are possible; if the key is a single letter there are only 25 keys.

3.2.2 Vigenère cipher

★ Asked in S2026 Q3(a) [3M: Vigenère KEEPMESSAGESECRET key CIPHER]

Vigenère cipher: polyalphabetic. A keyword is **repeated** under the plaintext; each letter is shifted by the key letter (A=0 ... Z=25).

Encryption: $C_i = (P_i + K_i) \text{ mod } 26$ **Decryption:** $P_i = (C_i - K_i) \text{ mod } 26$

Solved (S2026): "KEEP MESSAGE SECRET", key = CIPHER

Remove spaces → KEEPMESSAGESECRET (17 letters). Repeat key CIPHERCIPHERCIPHE...

Plain	K	E	E	P	M	E	S	S	A	G	E	S	E	C	R	E	T
P	10	4	4	15	12	4	18	18	0	6	4	18	4	2	17	4	19
Key	C	I	P	H	E	R	C	I	P	H	E	R	C	I	P	H	E
K	2	8	15	7	4	17	2	8	15	7	4	17	2	8	15	7	4
(P+K) mod 26	12	12	19	22	16	21	20	0	15	13	8	9	6	10	6	11	23
Cipher	M	M	T	W	Q	V	U	A	P	N	I	J	G	K	G	L	X

Ciphertext = MMTWQVUAPNIJGKGLX

Note: the Gujarati question says 'KEEP THE MESSAGE SECRET' but the English (authentic) version has no 'THE'.
If you write the Gujarati version: KEEPTHEMESSAGESECRET → MMTWXYGUTZWIRIMHLGIB.

3.2.2 Playfair cipher

★ Asked in W2025 Q3(b)OR [4M: Playfair, plaintext battery, key FORUM]

Playfair cipher: encrypts **pairs of letters (digraphs)** using a 5 × 5 key square (I and J share one cell).

Steps

1. Build the 5 × 5 grid: write the key (no repeated letters) first, then the remaining alphabet letters.
2. Split plaintext into pairs. If a pair has the **same letters**, insert **X** between them. If one letter is left alone, add **X**.
3. **Same row** → take the letter to the **right** of each (wrap around).
4. **Same column** → take the letter **below** each (wrap around).
5. **Rectangle** → each letter is replaced by the letter in **its own row** and the **other letter's column**.

Solved: plaintext BATTERY, key FORUM

Grid (key FORUM in first row, then A B C D E G H I K L N P Q S T V W X Y Z):

F	O	R	U	M
A	B	C	D	E
G	H	I	K	L
N	P	Q	S	T
V	W	X	Y	Z

Pairs: BA | TT → TX | TE | RY → **BA, TX, TE, RY**

Pair	Rule	Result
BA	Same row (2nd row): B → C, A → B	CB
TX	Rectangle: T (row 4, col 5), X (row 5, col 3) → Q, Z	QZ
TE	Same column (col 5): T → Z (below), E → L (below)	ZL
RY	Rectangle: R (row 1, col 3), Y (row 5, col 4) → U, X	UX

Ciphertext = CBQZZLUX

3.2.2 Hill cipher

★ Hill cipher asked in BOTH papers: W2025 Q3(c) [7M: decisions, 3×3] + S2026 Q3(b)OR [4M: DATA, 2×2]

Hill cipher: polyalphabetic cipher that encrypts a **block of letters** at once using **matrix multiplication** mod 26 (a=0 ... z=25).

Encryption: $C = (P \times K) \bmod 26$ Decryption: $P = (C \times K^{-1}) \bmod 26$

- P = row of plaintext numbers (block size = size of key matrix), K = key matrix.
- The key matrix must be **invertible mod 26**: $\gcd(\text{determinant}, 26) = 1$.
- If the last block is short, pad with a filler letter (x or z).

Solved (W2025): plaintext "decisions", key $K = \begin{bmatrix} 5 & 7 & 10 \\ 3 & 1 & 7 \\ 0 & 5 & 2 \end{bmatrix}$

Numbers: d=3, e=4, c=2, i=8, s=18, i=8, o=14, n=13, s=18. Blocks of 3: (d e c), (i s i), (o n s).

Check key: $\det(K) = 5(2-35) - 7(6-0) + 10(15-0) = -165 - 42 + 150 = -57 \equiv 21 \pmod{26}$.
 $\gcd(21, 26) = 1 \rightarrow$ invertible ✓

Block	Row $\times$ K (each column)	Mod 26	Letters
(3, 4, 2)	$3 \times 5 + 4 \times 3 + 2 \times 0 = 27$ $3 \times 7 + 4 \times 1 + 2 \times 5 = 35$ $3 \times 10 + 4 \times 7 + 2 \times 2 = 62$	1, 9, 10	BJK
(8, 18, 8)	$8 \times 5 + 18 \times 3 + 8 \times 0 = 94$ $8 \times 7 + 18 \times 1 + 8 \times 5 = 114$ $8 \times 10 + 18 \times 7 + 8 \times 2 = 222$	16, 10, 14	QKO
(14, 13, 18)	$14 \times 5 + 13 \times 3 + 18 \times 0 = 109$ $14 \times 7 + 13 \times 1 + 18 \times 5 = 201$ $14 \times 10 + 13 \times 7 + 18 \times 2 = 267$	5, 19, 7	FTH

Ciphertext = BJKQKOFTH

Solved (S2026): plaintext "DATA", key $K = \begin{bmatrix} 5 & 8 \\ 17 & 3 \end{bmatrix}$

Blocks of 2: (D A) = (3, 0) and (T A) = (19, 0). **Check key:** $\det = 5 \times 3 - 8 \times 17 = -121 \equiv 9 \pmod{26}$; $\gcd(9, 26) = 1$ ✓

Block	Row $\times$ K	Mod 26	Letters
(3, 0)	$3 \times 5 + 0 \times 17 = 15$ $3 \times 8 + 0 \times 3 = 24$	15, 24	PY
(19, 0)	$19 \times 5 + 0 \times 17 = 95$ $19 \times 8 + 0 \times 3 = 152$	17, 22	RW

Ciphertext = PYRW

Note: this uses the row-vector method $C = P \times K$ (as in Forouzan/Kahate and the W2025 paper). If your teacher uses the column method $C = K \times P$, the answer would be PZRL - mention your method clearly in the exam.

3.2.2 One-time pad cipher

★ Asked in W2025 Q5(b)OR [4M: one-time pad with example]

One-time pad (Vernam cipher): the key is a **truly random** string, **as long as the message**, and used **only once**. It gives **perfect secrecy** (unbreakable) because any plaintext of that length is equally possible.

Encryption: $C = (P + K) \bmod 26$ Decryption: $P = (C - K) \bmod 26$ (for binary data: $C = P \oplus K$).

Example: plaintext HELLO, random key XMCKL

Plain letter	H	E	L	L	O
P	7	4	11	11	14
Key letter	X	M	C	K	L
K	23	12	2	10	11
$(P + K) \bmod 26$	4	16	13	21	25
Cipher letter	E	Q	N	V	Z

Ciphertext = EQNVZ. Decryption: $E(4) - X(23) = -19 \equiv 7 = H$, and so on.

- **Advantages:** perfect secrecy, simple.
- **Problems:** key must be as long as message, truly random, never reused, and hard to distribute safely - so it is used only in very high-security cases.

3.3.1 Keyless transposition: Rail fence, Rectangular (columnar)

Transposition cipher: letters are **rearranged** (positions change), not replaced.

1) Rail fence (keyless)

Write the plaintext diagonally in zig-zag over a number of 'rails', then read rail by rail. Example HELLOWORLD, 2 rails:

```
H   L   O   O   L           rail 1
  E   L   W   R   D           rail 2
```

Ciphertext = **HLOOL ELWRD** (rail 1 then rail 2).

2) Rectangular / columnar (keyless)

Write plaintext row by row in a fixed number of columns, read **column by column** (left to right). Example MEETMEAFTERTHEEXAM, 4 columns:

```
M E E T
M E A F
T E R T
H E E X
A M
```

Reading columns: MMTHA + EEEEM + EARE + TFTX → **MMTHAEEEEEMEARETFTX**.

3.3.2 Keyed transposition

★ Asked in W2025 Q3(a)OR [3M: keyed transposition with example]

Keyed transposition cipher: plaintext is divided into **blocks** (size = key length) and the letters of each block are **permuted** according to a secret key (a permutation).

Example: key = (3 1 4 2) means plaintext letter in position 1 moves to position 3, position 2 → 1, position 3 → 4, position 4 → 2.

Plain block	H E L P
Key (position moves to)	1→3, 2→1, 3→4, 4→2
Cipher block	E P H L

Ciphertext = EPHL. Longer messages are split into blocks of 4 (pad the last block with a dummy letter, e.g. X) and each block is permuted the same way.

Decryption uses the inverse key: swap the roles of positions and values. Inverse of (3 1 4 2) is (2 4 1 3). Applying it to EPHL gives back HELP ✓.

3.3.2 Keyed columnar transposition and key inversion

★ Asked in S2026 Q3(c) [7M: keyed columnar, MEETMEAFTERTHEEXAM, key 4312]

Keyed columnar transposition: write plaintext **row-wise** in a grid with as many columns as key digits; then read **column-wise in the order given by the key** (column marked 1 first, then 2, ...).

Solved: plaintext MEETMEAFTERTHEEXAM, key = 4 3 1 2

Plaintext has 18 letters; with 4 columns we get 5 rows (last row has 2 letters).

Key →	4	3	1	2
Row 1	M	E	E	T
Row 2	M	E	A	F
Row 3	T	E	R	T
Row 4	H	E	E	X
Row 5	A	M		

- Column with key 1 (3rd column): **E A R E**
- Column with key 2 (4th column): **T F T X**
- Column with key 3 (2nd column): **E E E E M**
- Column with key 4 (1st column): **M M T H A**

Ciphertext = EARE TFTX EEEEM MMTHA = EARETFTXEEEEEMMMTHA

Key inversion (for decryption): key 4 3 1 2 says column 1 is read 4th, column 2 is read 3rd, column 3 read 1st, column 4 read 2nd. The inverse key is **3 4 2 1**: the 1st block of ciphertext goes to column 3, the 2nd to column 4, the 3rd to column 2, the 4th to column 1. Fill these back and read row by row to get the plaintext.

Assumption: last row is left incomplete (no padding). If your teacher pads with X, say 'assuming padding X' - GTU allows suitable assumptions.

3.3.3 Stream and Block ciphers

★ Asked in W2025 Q3(a) [3M: stream cipher and block cipher with example]

Point	Stream cipher	Block cipher
Processing	Encrypts one bit / one byte at a time with a keystream (usually XOR)	Encrypts a fixed-size block (e.g. 64 or 128 bits) at a time
Speed / use	Fast, good for real-time data (voice, video, Wi-Fi)	Slower, good for files and bulk data
Error effect	A bit error affects only that bit	An error can affect the whole block
Key	Key stream generated from a key (one-time pad is the ideal stream cipher)	Same key used for each block
Examples	RC4, one-time pad, Vigenère (letter by letter)	DES (64-bit), AES (128-bit), Hill cipher, Playfair (pairs)

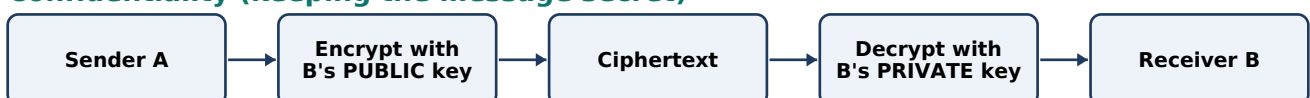
3.4.1-3.4.2 Asymmetric-key cryptosystem: ingredients, confidentiality, authentication

Asymmetric (public-key) cryptography uses **two related keys**: a **public key** (shared with everyone) and a **private key** (kept secret). What one key encrypts, only the other can decrypt.

Ingredients of public-key encryption (five)

1. Plaintext
2. Encryption algorithm
3. Public key and private key (a pair)
4. Ciphertext
5. Decryption algorithm

Confidentiality (keeping the message secret)



Only B has the private key, so only B can read the message.

Authentication (proving who sent it - digital signature)



Only A has A's private key, so a message that opens with A's public key must have come from A.
(Both together: encrypt first with A's private key, then with B's public key = authentication + confidentiality.)

Point	Symmetric	Asymmetric
Keys	1 shared secret key	2 keys: public + private
Speed	Fast	Slow
Key distribution	Difficult	Easy (public key is open)
Examples	DES, AES	RSA, ECC

3.4.3 RSA algorithm and key generation

★ **RSA asked in BOTH papers (7M each): W2025 Q3(c)OR [key generation + enc/dec with example] + S2026 Q3(c)OR [p=5, q=11, e=7, M=4]**

RSA (Rivest-Shamir-Adleman) is the most popular public-key algorithm. Its security depends on the difficulty of **factoring a large number n into its primes p and q**.

Key generation

1. Choose two large primes **p** and **q**.
2. Calculate **n = p × q**.
3. Calculate **φ(n) = (p - 1)(q - 1)**.
4. Choose **e** with $1 < e < \phi(n)$ and **gcd(e, φ(n)) = 1**.
5. Find **d** such that **(d × e) mod φ(n) = 1** (use Extended Euclid).
6. **Public key = {e, n}**, **Private key = {d, n}**.

Encryption: $C = M^e \bmod n$ Decryption: $M = C^d \bmod n$ (M must be less than n).

Solved example: p = 5, q = 11, e = 7, M = 4

Step	Calculation
n	$5 \times 11 = 55$
φ(n)	$(5 - 1)(11 - 1) = 4 \times 10 = 40$
e	7 (gcd(7, 40) = 1 ✓)
d	$7 \times d \equiv 1 \pmod{40} \rightarrow d = 23$, because $7 \times 23 = 161 = 4 \times 40 + 1$
Keys	Public = {7, 55}; Private = {23, 55}
Encryption	$C = 4^7 \bmod 55$. $4^2 = 16$; $4^4 = 256 \bmod 55 = 36$; $4^7 = 4^4 \times 4^2 \times 4 = 36 \times 16 \times 4 \rightarrow (576 \bmod 55 = 26) \rightarrow 26 \times 4 = 104 \bmod 55 = \mathbf{49}$
Decryption	$M = 49^{23} \bmod 55$. $49^2 \bmod 55 = 36$; $49^4 = 36^2 = 1296 \bmod 55 = 31$; $49^8 = 31^2 = 961 \bmod 55 = 26$; $49^{16} = 26^2 = 676 \bmod 55 = 16$. Since $23 = 16 + 4 + 2 + 1$: $16 \times 31 \times 36 \times 49 \bmod 55 = (496 \bmod 55 = 1) \rightarrow 1 \times 36 = 36 \rightarrow 36 \times 49 = 1764 \bmod 55 = \mathbf{4}$

Decrypted message = 4 = original M ✓.

Exam tip: In exams just show the steps (n, φ, d, C, M). If your calculator cannot do large powers, use the square-and-multiply method shown above.

Last-minute revision sheet - Unit 3

- Letters: $a=0 \dots z=25$, everything mod 26.
- Additive: $C = (P+K) \bmod 26$. Multiplicative: $C = (P \times K) \bmod 26$ (K co-prime with 26). Affine: $C = (aP+b) \bmod 26$.
- Vigenère: repeat the keyword, add letter values. Autokey: key + plaintext itself. OTP: random key, same length, used once.
- Playfair: 5×5 grid, pairs; same row $\rightarrow$ right, same column $\rightarrow$ down, rectangle $\rightarrow$ swap columns; double letter $\rightarrow$ insert X.
- Hill: $C = P \times K \bmod 26$; decrypt with K^{-1} ; $\det(K)$ must be co-prime with 26.
- Columnar: write in rows, read columns (key order for keyed). Decrypt using inverse key.
- RSA: $n = p \times q$, $\phi = (p-1)(q-1)$, e co-prime to ϕ , $d \times e \equiv 1 \pmod{\phi}$. $C = M^e \bmod n$, $M = C^d \bmod n$.
- Public key: encrypt with receiver's PUBLIC key = confidentiality; encrypt with sender's PRIVATE key = authentication.